

TITLE PAGE

• **Problem Statement ID –** 26176

• **Problem Statement Title -** ORCA Marine EcOsystem Reasoning
with Collaborative Agents

• **Theme-** Disaster Management

• **PS Category-** Software

• **Team ID-** 177481

• **Team Name -** BitByBits

Website Link- [Link](#)

Website Demo- [Link](#)

Github Repo- [Link](#)

Report Link- [Link](#)

1 Problem Statement

Marine decisions are data rich but intelligence poor



Critical **information** across weather, oceans, fisheries, safety and geography is **scattered**, making fast decisions **difficult**.



Fragmented & Siloed Data



High cognitive load



Time Critical Safety Risk



Generic Advisories

India's Revised Coastline → **11,098.81 km**

Fishers & Fish Farmers → **2.8 crore+**

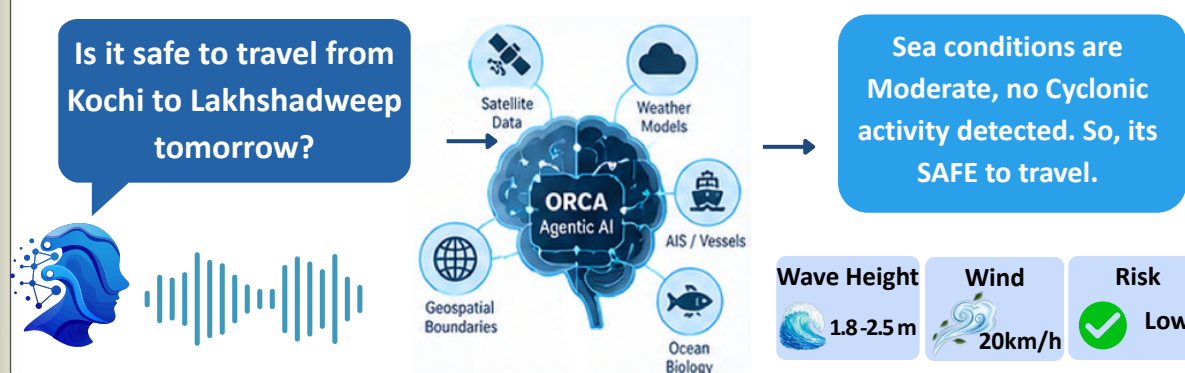
Marine Capture Fisheries Production → **44.95 lakh tonnes**

2 Gap Analysis

No single intelligence layer

Existing Approach	Missing Capabilities
Data portals and Dashboards	Natural Language interaction
Individual APIs	cross source evidence fusion
Layer/map viewers	context aware reasoning
Static forecasts	Multi model risk inference

How ORCA Works



3 Our Methodology

- 1 Understand Context** Intent, voice (English, Hindi and 9 other coastal languages), coordinates, vessel class (LOA).
- 2 Plan Autonomously** LangGraph multi-agent DAG decomposes queries into executable tasks.
- 3 Discover & fuse** 3 Live satellites (NOAA SST/Chl), CMEMS currents, Open-Meteo, AIS & EEZ WFS.
- 4 Reason in space & time** Uber H3 hexagonal indexing (Res-6), thermal fronts (SIED), geofencing.
- 5 Deterministic Safety & AI** Wave, convection, visibility, collision, overall risk output by deep learning model.
- 6 Explain, Route & Act** A safe passage planning, multilingual voice synthesis, CAP-1.2 XML alerts.

4 Why ORCA?

One intelligence layer.
Multiple marine decisions.



Fishers

Vessel-specific safety decisions with reasoning



Researchers

Stronger research with trusted data and personalized datasets



Accurate Forecasting

Forecast anomalies with Deep Learning based model



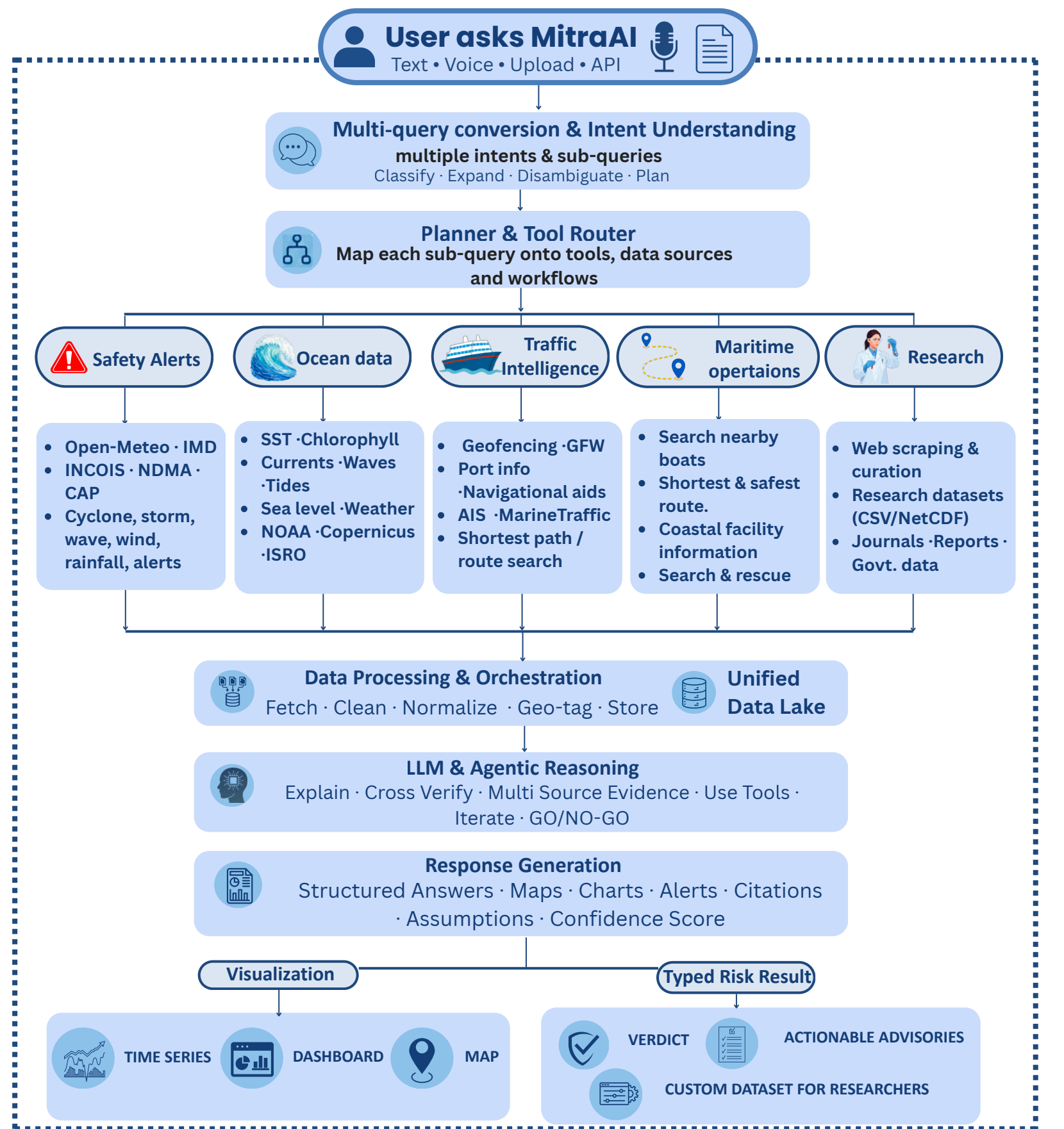
Disaster Management

Multilingual warnings beyond internet coverage



Maritime Operators

Quick alert, focused search zone, efficient operation.

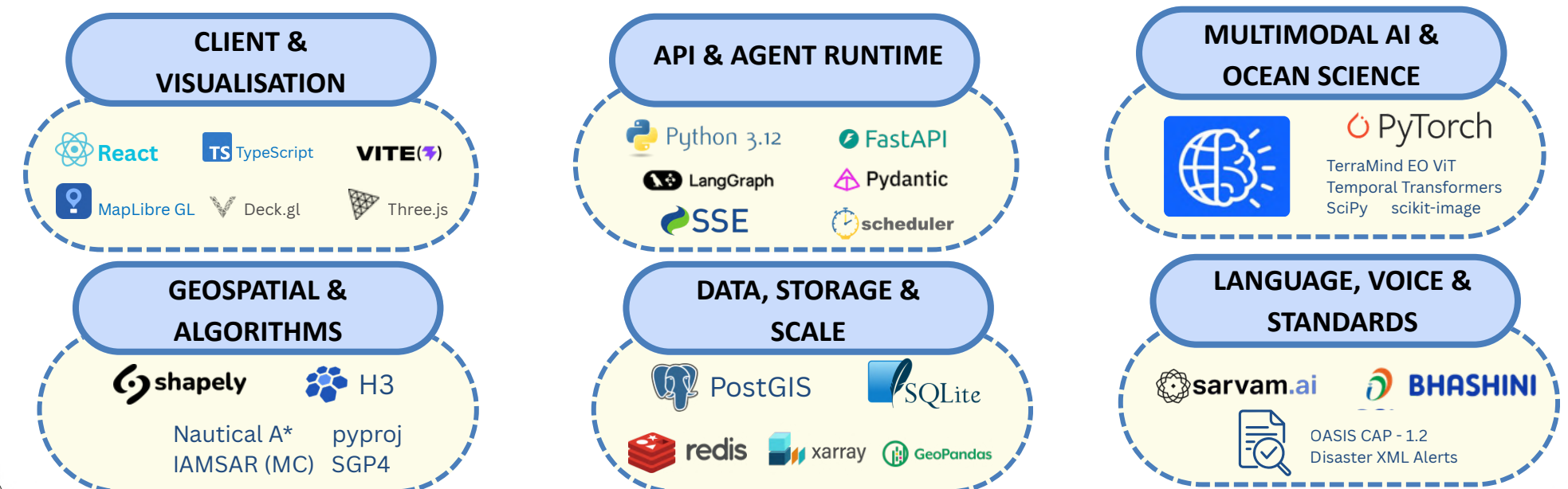


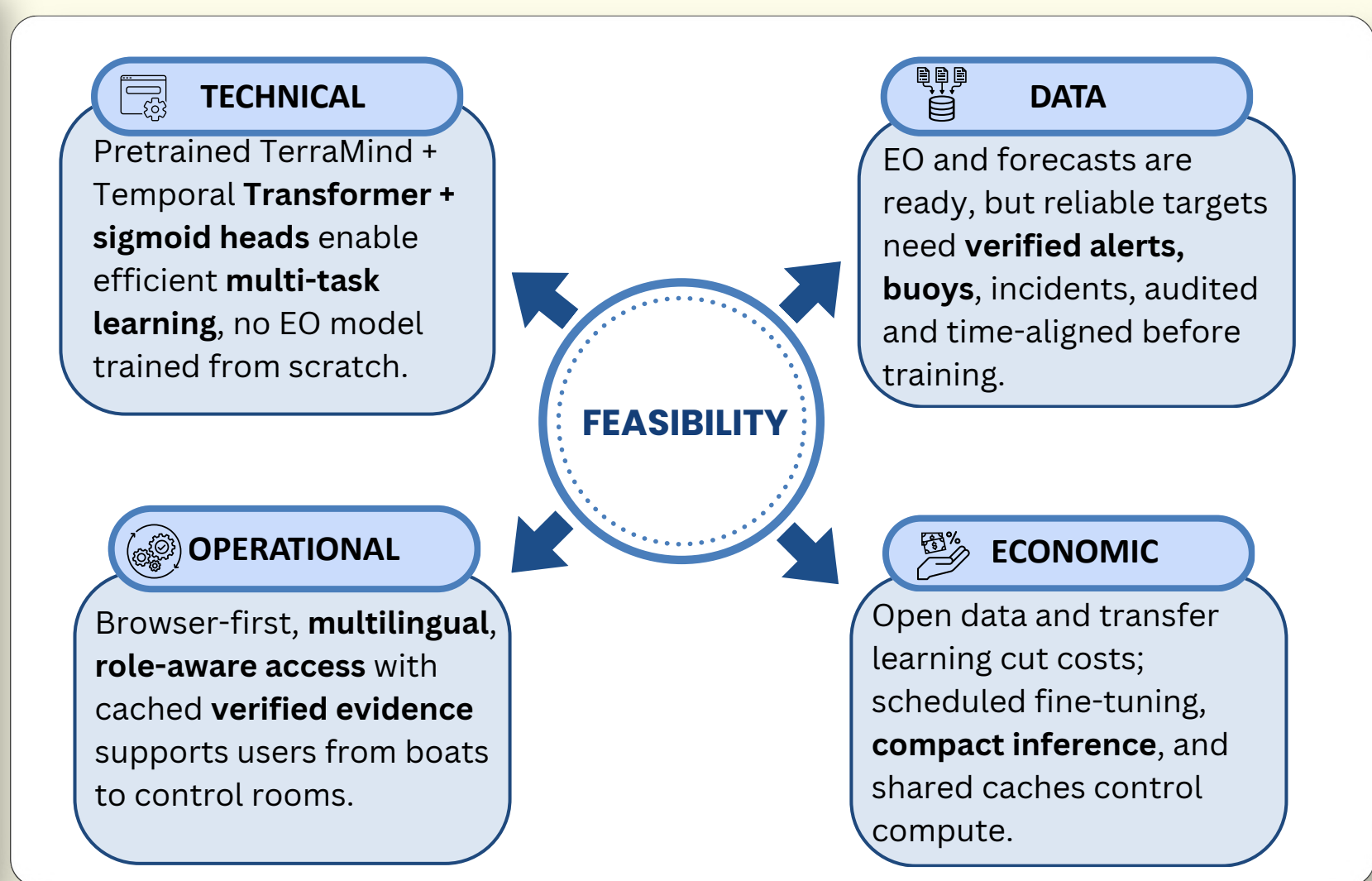
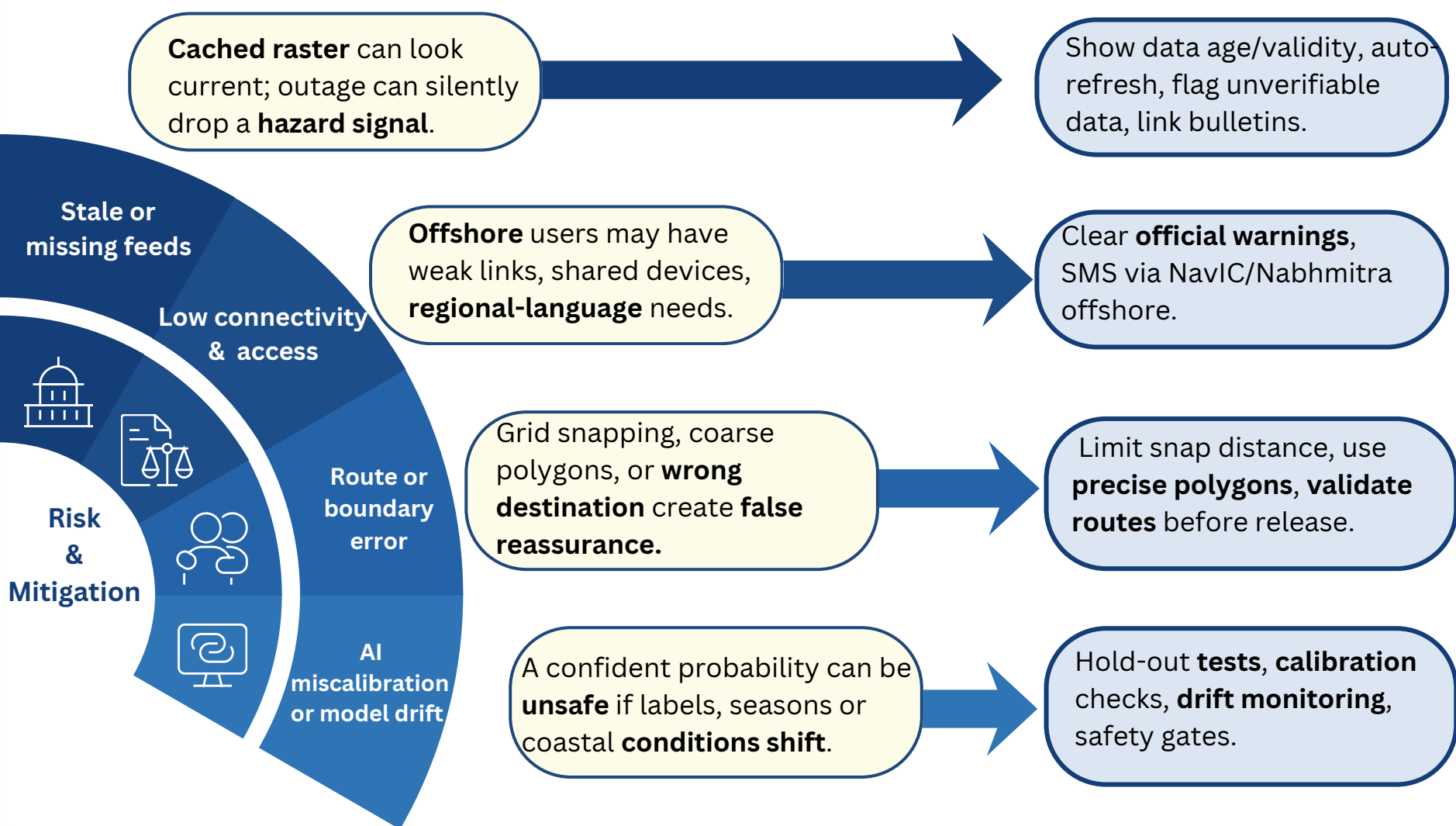
DATA AND API SOURCES

SOURCE/API	ROLE IN ORCA
Open Meteo Marine + Forecast	Waves, swell, wind, gusts, visibility, CAPE at clicked point. Hourly updates, ~20 min cache.
IMD advisories + INCOIS ERDDAP	Official fishermen/cyclone/port warnings. SST, chlorophyll.
JPL MUR via NOAA ERDDAP	Satellite SST for thermal fronts & PFZ. Refreshed every 24h.
ESA CCI chlorophyll • Copernicus / NASA / Sentinel	Monthly chlorophyll + SST for PFZ. Others used as optional cross-checks.
Marine Regions • EEZ / IMLB	EEZ & maritime boundaries for jurisdiction (static file, not weather).
Celestrak TLE • AISStream • GFW	Next satellite pass, nearby ships (AIS), public fishing-activity map.

SIGNIFICANT WAVE HEIGHT 1.08 m LIVE Open-Meteo • 13 min ago	PEAK PERIOD 11.6 s LIVE Open-Meteo • 13 min ago
SWELL HEIGHT 0.96 m LIVE Open-Meteo • 13 min ago	WAVE DIRECTION 175.0 deg LIVE Open-Meteo • 13 min ago
WIND SPEED 5.10 kn LIVE Open-Meteo • 13 min ago	GUSTS 8.00 kn LIVE Open-Meteo • 13 min ago
WIND DIRECTION 113.0 deg LIVE Open-Meteo • 13 min ago	VISIBILITY 15260 m LIVE Open-Meteo • 13 min ago
CAPE (INSTABILITY INDICAT... 1890 J/kg LIVE Open-Meteo • 13 min ago	CIN 4.00 J/kg LIVE Open-Meteo • 13 min ago
PRECIPITATION 0.00 mm LIVE Open-Meteo • 13 min ago	SEA SURFACE TEMP (MODEL) 30.9 degC LIVE Open-Meteo • 13 min ago

TECH STACK

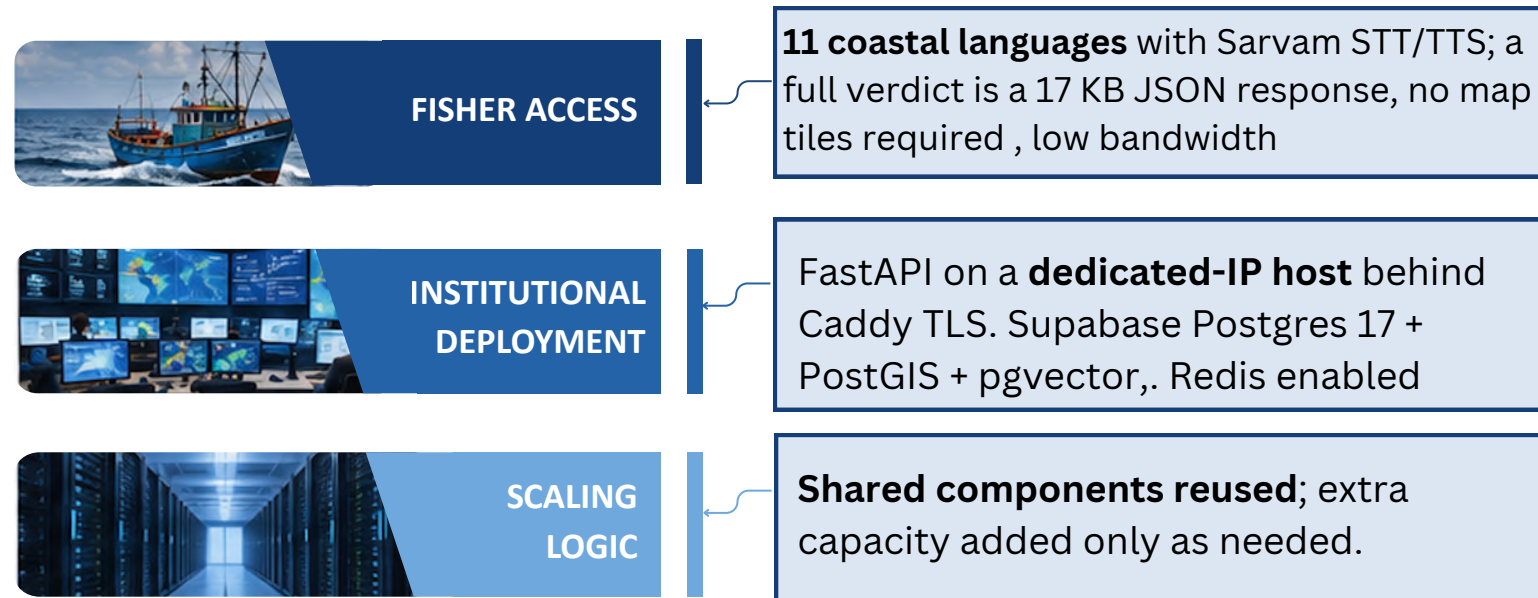




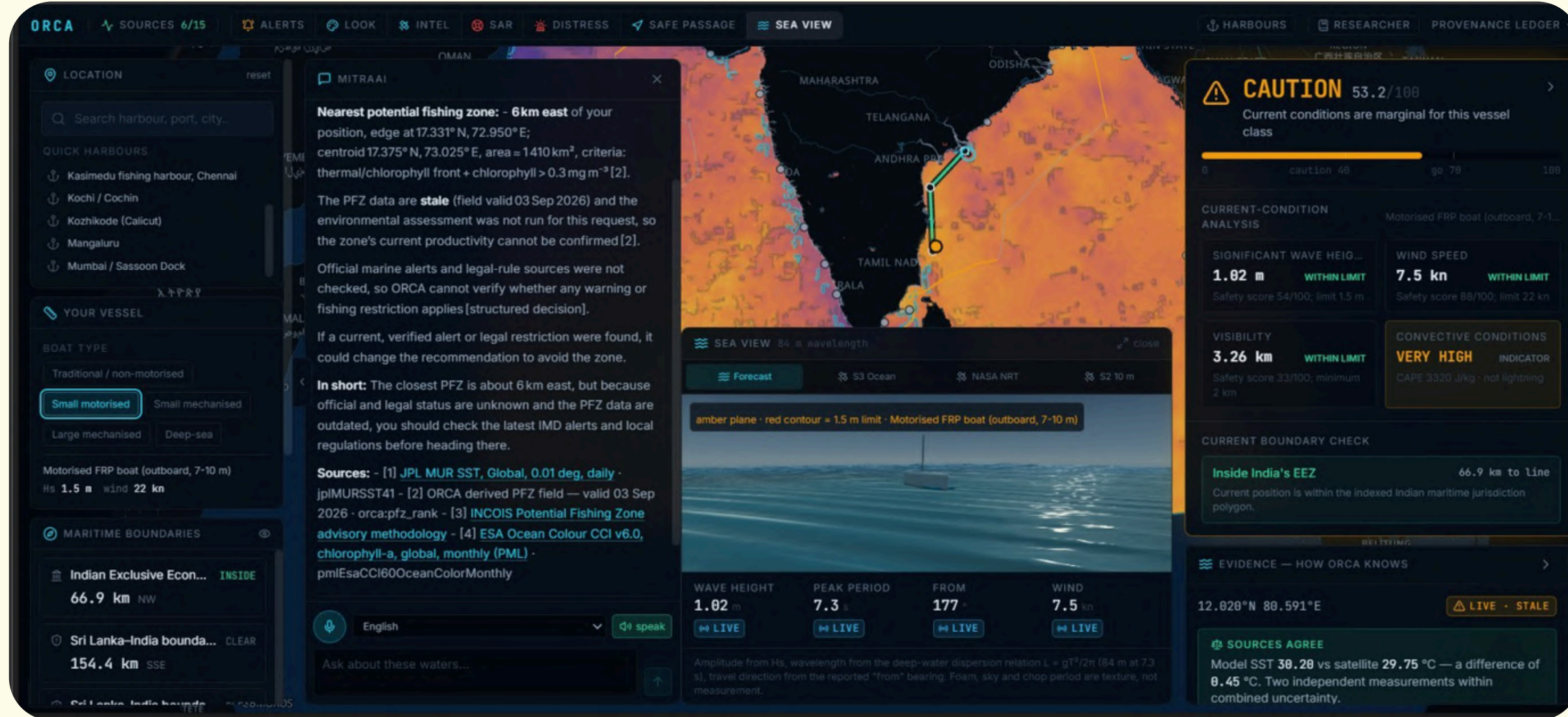
Sample Dataset for researchers along with pretrained models available for Fine Tuning

date	latitude	longitude	point	sst	wave_height	wind_speed	wave_direction
2026-08-20	10.5	75.75	centre	26.575	1.8	21.6	239
2026-08-21	10.5	75.75	centre	26.624	1.64	25.2	239
2026-08-22	10.5	75.75	centre	27.318	1.9	21.8	242
2026-08-23	10.5	75.75	centre	27.627	1.7	30	254
2026-08-24	10.5	75.75	centre	27.462	1.7	28.3	255
2026-08-25	10.5	75.75	centre	26.57	1.64	28.3	256
2026-08-26	10.5	75.75	centre	26.586	1.54	21.7	253
2026-08-27	10.5	75.75	centre	26.63	1.4	17.5	252

VIABLE OPERATING MODEL



1.



3.

EAST COAST			Kanyakumari to the Sundarbans · 10 stretches
NO-GO	Pazhayar → Tharuvaikulam	2 harbours	
CAUTION	Vellapallam → Arcottuthurai	2 harbours	
GO	Karaikal → Tharangambadi	2 harbours	
CAUTION	Thengaithittu → Thiruvottiyur Kuppam	3 harbours	
GO	Juvvaladinne → Vodarevu	3 harbours	
CAUTION	Nizampatnam	1 harbour	
GO	Machilipatnam → Uppada	2 harbours	
NO-GO	Pudimadaka → Petuaghat	8 harbours	

Website Link

1. Dashboard :

It is a snapshot of the dashboard. It shows the whole **interactive graph with SST** and the scoring via multiple variables for each point on the right side. **MitraAI answers** the questions with relevant **sources** and **citations**. On the left side, it shows **boat types**, **India's EEZ and boundaries**, and some seaside locations. In the Sea View section, we show recent images of a particular point from **3 different satellites** and also a **3D digital twin**.

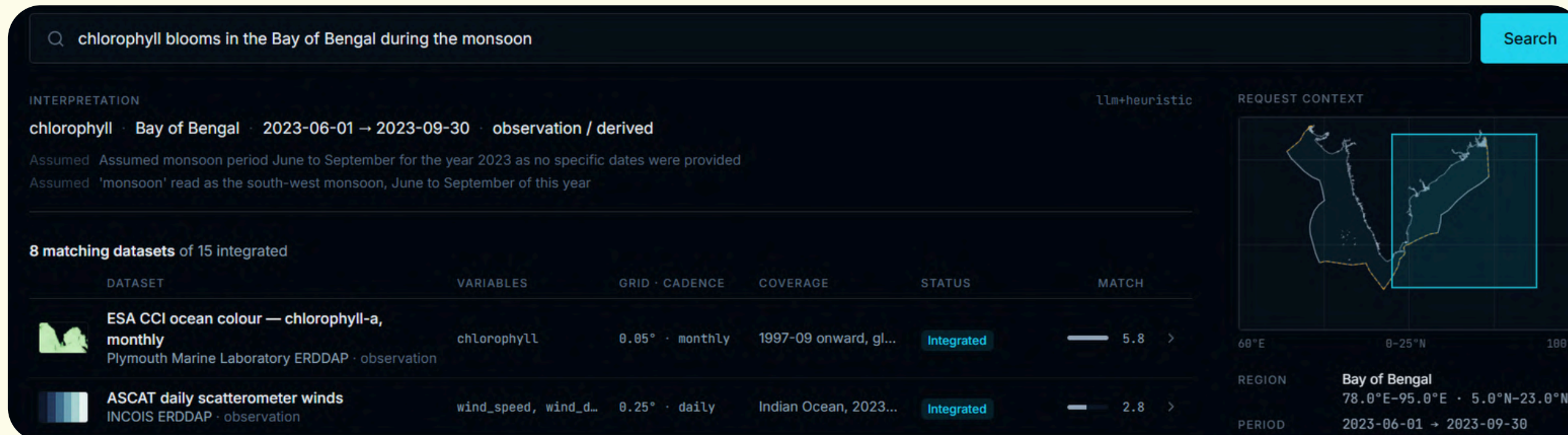
2. Researcher :

Here, the researcher can simply **prompt in natural language** what type of **data**, for which **coastal regions**, and from which **date**. Our agent performs a **deep web search** and **fetches data** from multiple open source sources. After **cleaning and preprocessing** the messy data, it creates a **clean dataset** file for the **researcher**.

3. Coastal Authorities :

Authorities can see the condition of **every harbour at a glance** and operate and **manage the entire coastal boundary** through our system.

2.



FUTURE ASPECT

LAST-MILE SAFETY

Offline Multilingual Alert Relay



Satellite Message Relay



Multilingual Alerts (Tamil/Hindi/Malayam)



Works Offline (SMS / NavIC)



Verified by IMD/INCOIS/NDMA

RESEARCH AT CONVERSATIONAL SPEED

Researcher Marine Data Cube

Ask spatial temporal questions without manually reconciling portals, grids and timestamps.

Co-register EO, model & buoy data; detect anomalies, trends and time lags; export GeoTIFF, NetCDF or CSV with query & data lineage.



LEARNED MULTI-HAZARD INTELLIGENCE

Multi-Output Probability Model



SST



Wind & Waves



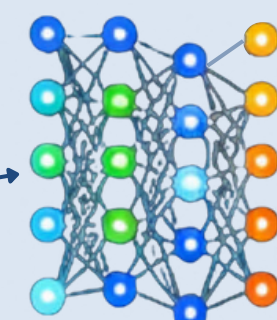
Vessel Traffic



AIS



Historical Data



Anomaly Detection



Cyclone Prediction



Fishing Zone (PFZ)



Wave Forecast



Risk Scoring

BENEFITS

AUDIENCE	DECISION TRANSFORMED	PRACTICAL VALUE
FISHERS	High Wave Alert in 2 hours	Safer go/no-go decisions, faster planning, route awareness..
MARINE RESEARCHERS	MANUAL DOWNLOADS → REPRODUCIBLE INSIGHTS	Faster hypothesis screening, source comparison, & various datasets
COASTAL AUTHORITIES	Unified situational view	Prioritised monitoring, clearer communication.
DISASTER MANAGEMENT	STATIC PRODUCTS → TIMELY TARGETED ANSWERS	Situational awareness, targeted outreach, better coordination.
MARITIME OPERATORS	Optimized Route	Lower hazard exposure, defensible routes, operational hand-offs.

Research and References

- Small-craft operability / Hs limits — J. Mar. Sci. Eng. (2023) <https://doi.org/10.3390/jmse11071302>
- Development of small vessel advisory and forecast services system - <https://www.tandfonline.com/doi/full/10.1080/1755876X.2020.1846267>
- IMD marine warnings & API products (<https://mausam.imd.gov.in/>)
- TerraMind: Large-Scale Generative Multimodality for Earth Observation - <https://arxiv.org/abs/2504.11171>